

DOUBLE JEOPARDY: The Vulnerability Spiral — Quantifying Compound Coastal Risk and Governance Evidence Alignment Across Five Island Nations

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Entry 1

Project Overview

I Am running the DOUBLE JEOPARDY project to check a big danger. (See, the main issue is tht rising sea levels hit small islands badly because their natural walls like trees and rocks are dying). I did not want to put all coastal green stuff into jst one big group. So, I chose to track mangroves and coral reefs completely separately. My data comes from five different island nations across three big ocean basins: Maldives, Lakshadweep, Seychelles, Fiji, and the Canary Islands. Calculated the health paths for corals and mangroves as two different things instead of mixing them into one single score. Finally, I mixed these separate numbers with the real physical danger data to make one big final score.

I am also looking at governance. (Basically, I want to see if officially making a place a protected area—which is wht big global groups and island leaders always say they are doing to control the danger to the beach—actually matches up with the spots where my numbers show the highest real danger). Chose to check this because it moves my work away from jst basic science. Am testing if government rules actually follow the real field proof or if they are jst totally disconnected from each other. My findings will matter a lot for things like adaptation finance, environmental economics, and marine spatial governance.

Problem Statement

Big money groups and climate policy makers think all beach ecosystems are jst one single safety category. They assume tht mangroves and coral reefs will always die together at the same time and jointly decide how badly a village gets hit by sea level rise, but nobody really tests this old idea with proper independent data over long timelines on a proper big cross national scale. So I think this is a huge mistake..... (If these two items are actually dying at different speeds or going in totally opposite directions, then putting them in one

single bucket will make leaders spend money in the wrong places and completely miscalculate which coastal zones are in real trouble).

Then there is a 2nd major issue on top of that. Global cash frameworks and national laws use the "protected area" tag as a fake proof that the ecosystem is being managed nicely..... (In reality, people draw these protection borders because of old politics or administrative ease instead of checking new, live risk numbers). I found that nobody has used independent satellite evidence before my project to check if these five islands have real safety or just "paper parks" that are totally disconnected from the real danger zones. So I wanted to see if the protection lines actually match the highest ecological and physical vulnerability.....

Aim

I am building a reproducible geospatial framework. (My main goal here is to separately count the actual physical danger from sea level rise and track how mangroves and coral reefs are changing over multiple time periods across five different island nations so I can finally create a real, evidence based composite vulnerability assessment). Then I will do a 2nd step. I will run stats to check if the current protected area governance lines actually match up with the real danger spots I found, or if there is a massive governance evidence gap happening on the ground.

Research Questions

My 1st question is about nature..... (Basically, I am trying to find out if the real physical danger of sea level rise mixes up with dying natural borders to create a much worse total danger for villages across five island nations, and if this whole process looks exactly identical for every ecosystem type or if it actually depends on whether mangroves or corals are the ones rotting away in that specific spot). So I need to solve this doubt 1st.

Then I have my 2nd question. Want to check if the actual places where governments put official WDPA protected area tags match my field data on high danger, or if these official marks are just completely independent of—or maybe even opposite to—the real ecosystem risks I found using satellite maps.

Hypotheses

My 1st guess is about coral reefs. (Think my long term stats will show a clear increase in thermal bleaching stress over many decades, especially on islands that face much higher physical sea level rise exposure). Will check this 1st.

Next, I look at mangroves. I think my data will show they are shrinking in total area over this exact same time frame as a totally separate problem alongside the coral dying. (I am going to test this idea just as strictly as the 1st one, and I will write down the truth in my log no matter what my final charts actually show).

Finally, I have a governance guess. Expect that islands and coastlines with high total vulnerability scores will have more protected area coverage because rulers should react to real danger. But if I find zero connection or the complete opposite thing, that is still a real result proving that protection rules in my sample are not based on actual risk data, and I will report it honestly either way.....

Objectives

1st, I will build a physical exposure dataset for villages across five island nations. (Am using basic height and slope map data to calculate exactly how many houses in local settlements sit below the dangerous one meter sea level rise mark). This is my starting point.

Then I will download the full Global Mangrove Watch archive. I chose to track real area changes across several separate years instead of just doing a lazy before and after comparison. Next, I need to check the coral reefs. (I am using satellite Degree Heating Week data to map out if bleaching heat stress is actually rising over time).

After that, I will mix everything. I am combining my physical danger maps and ecosystem loss data to calculate a single Compound Vulnerability Score for every island..... Then I am going to check the official WDPA protected area maps. Will run a stats test to see if these government lines actually match my final vulnerability scores.....

I also want to track new buildings..... (I will use satellite maps to do built up area change detection to see if houses and infrastructure are expanding right inside or next to the fragile ecosystem buffer zones). Am going to double check every single trend using multiple independent time points so my data stays clean. Finally, I will package all these scripts into a reproducible geospatial framework in an interactive dashboard to show my risk findings in governance analysis clearly to adaptation policy and conservation finance people.

Methodology Summary

Figured out physical exposure 1st. (Basically, I took the height and slope map data and crossed it with where people actually live on all five islands so I could calculate the percentage of houses sitting right at or below the one meter line). This gave me the main numbers.

Then I looked at the nature data. For mangroves, I chose to use multiple independent time points from the Global Mangrove Watch archive because I wanted to check for real area changes over a few decades. Then I checked the corals..... (Took a continuous multi decadal satellite thermal stress time series and simply compared the old averages against the recent ones to see how bad things got).....

Finally, I mixed the three data tracks. So I combined the physical exposure numbers with my two separate ecosystem signals to calculate a single, normalized Compound Vulnerability Score for each island.

Next, I am coding the governance part..... (So I counted the total size and density of official protected areas using data from the World Database on Protected Areas for every island and even for single coastal zones where the map resolution was good enough). Ran a stats test after that..... I directly compared those protection numbers against my final Compound Vulnerability Score because I want to see if high risk zones are more likely, equally likely, or less likely to get official government protection. This step is very important to me. (Instead of just writing generic comments about politics at the very end of my paper, I chose to convert my nature findings into a real, testable statement to check if government policies actually work on the ground).

So I will do a later phase. (For this part, I plan to use satellite based built up area change detection to map out if new houses, roads, and village expansions are heavily building up right inside or close to the fragile ecosystem buffer zones). So I want to test a weird loop..... Think humans reacting to environment pressures might actually end up making the exact same vulnerability worse by destroying their own natural shields. Then I have one more check. (So I am going to calculate if this bad building encroachment happens more often inside official protected boundaries or completely outside of them).....

Study Area

I picked five island nations. (My list spreads across three ocean basins: Maldives and Lakshadweep in the Indian Ocean, Seychelles in the western Indian Ocean, Fiji in the South Pacific, and the Canary Islands over in the eastern Atlantic). So I had to choose these specific places because they have good, matching data from multiple sources. It helps my study. (Wanted massive geographic, geological, and climatic differences so my

compound vulnerability score nd governance alignment framework get tested across totally different island styles, law systems, nd administration types instead of jst checking one single boring case). This makes my data look solid.

Expected Outputs

So I hve specific expected outputs. (1st, I built a settlement level physical exposure dataset for villages across all five island nations). I also created a validated multi temporal mangrove extent dataset. Then I coded a validated multi decadal coral reef thermal stress dataset using satellite histories. (After that, I mixed these parts to generate a single composite Compound Vulnerability Score nd risk ranking across all five islands). This is the core data.

Next comes my political data. So I ran a governance alignment analysis to test the statistical relationship between protected area coverage and verified vulnerability..... (So I am specifically searching for "governance evidence gap" zones where my maps show high verified risk but the governments gave low or no formal protection). So I want to see the gaps.

Then I ran an expansion check. I created a settlement and infrastructure encroachment analysis to test the ecosystem buffer feedback hypothesis..... (I chose to break down these house building trends strictly by protection status to see where lines are failing). Finally, I packaged everything together. I built an interactive geospatial dashboard n an open source codebase tht other students can easily extend to more islands nd completely new ecosystem types.

Relevance

My project links to real world policy..... (I am talking directly to people working on climate adaptation policy, nature based solutions financing, ecosystem services valuation, nd marine spatial governance)..... I am contributing two main things. 1st, I found that treating coastal ecosystem protection as one big mixed category hides totally different risk realities on the ground, so doing strict, ecosystem specific, multi temporal verification matters way more than just believing some lazy uniform decline story..... This is a big eye opener. 2nd, I actually chose to test—instead of jst assuming blindly—whether current conservation governance tools line up with real, verified risk numbers. (By doing this, my dataset gives a fully transferable method so tht anyone can evaluate adaptation finance nd protected area prioritization choices in any coastal or island spot facing compound climate vulnerability). Think this will help a lot.

Entry 2

Status

Complete. Covers the initial project structure decisions made once the research question n five island study area were locked in, before any data acquisition started.

Starting Point

I fixed my research question. (Basically, I want to check if the real physical danger of sea level rise mixes with dying natural shields across five islands, n if this varies by ecosystem type). Then I had to make my 1st big decision. So I had to build a project folder structure that actually fits my study needs instead of jst copying some generic template from the internet. This was very necessary.

Folder Structure: Category Based, Not Island Based

So I had two choices before writing any data acquisition code. Could organize by island (meaning a separate Maldives folder, a Seychelles folder, with each place holding all its own data files), or I could organize strictly by data category (a settlements folder, a mangroves folder, with each holding all five islands' files together).

Chose the category based plan. (The main reason was my analysis methodology, because I need to compare the exact same variable across all five islands at once—like coral bleaching stress or encroachment rates—so an island based structure would mean opening five separate folders every single time a cross island script runs). It becomes a huge headache. Also, people easily make inconsistent naming mistakes when files are hidden inside separate island folders. Category based keeps every version of a dataset right in one place. My cross island script just runs a simple loop over the island names reading from the single folder. (It is immediately obvious if any island's file is missing from a category because all five files should be sitting right next to each other in the directory). This layout keeps my data clean.

The top level structure ended up as:

```
DOUBLE_JEOPARDY/
├── data/
│   ├── settlements/
│   ├── ecosystem_buffers/      (mangroves, coral reefs, protected areas)
│   ├── terrain/               (elevation, slope)
│   ├── population/
│   ├── cyclone_tracks/
│   ├── boundaries/           (island outlines)
│   ├── mangroves/            (raw multi-temporal GMW archives, kept separate
from processed ecosystem_buffers)
│   ├── coral_bleaching/      (NOAA thermal stress time series)
│   └── settlement_encroachment/ (satellite-derived built-up change data)
├── outputs/
│   └── plots/
├── DJ_Research_Paper.md
├── DJ_Executive_Summary.md
└── DJ_Development_Log.md
```

One Deliberate Inconsistency

So I made one deliberate inconsistency in my folder setup..... (The raw multi temporal Global Mangrove Watch archives for the 1996, 2010, n 2020 snapshots live inside their own mangroves/ folder, which is completely separate from the clean per island files inside the ecosystem_buffers/ directory). Chose this for practical reasons. The raw GMW shapefiles are absolutely massive since they weigh around 900 to 950 megabytes per year n cover the whole globe. Only need to process them once into tiny per island n per year extracts. (By keeping the giant source archives physically away from the small final files, I easily excluded the heavy raw archives from my Git version control while still tracking the lightweight output files nicely). This saved me from scattering folder level exceptions everywhere.....

Environment Setup

I reused my old gpie2 conda environment. (I did not want to start fresh because this specific environment was already fixed for a nasty numerical computing issue where a corrupted Intel MKL backend was breaking my scripts). So I solved tht old error by installing NumPy and SciPy with the nomkl OpenBLAS backend instead. Reusing it meant I did not risk facing tht same failure mode again in a clean environment. Two of my old files carried straight over to this project. I moved auth_sentinelhub.py for the Sentinel Hub OAuth module n my confidential .env file containing the Copernicus Data Space credentials. (So I did this because the API access patterns nd login credentials were 100% identical to wht I need for my new Sentinel-2 imagery work anyway). This saved a lot of setup time.

Why This Mattered Later

My upfront folder structure paid off beautifully. (Once I started the multi temporal analysis, everything became super easy because every mangrove year file, every coral time series file, nd every island's settlement file sat in a totally predictable, category consistent spot in my directory layout). This choice saved me hours of extra coding later..... My scripts looping across all five islands—like the elevation exposure checks, mangrove area comparisons, nd coral stress trends—could be written in a generic way. (Only had to change the island name between iterations instead of writing five near duplicate scripts hardcoded to five completely different folder setups). This trick kept my development log clean.

Entry 3

Status

So I finished this part completely. (My logs here cover how I pulled down and cleaned the three OpenStreetMap derived datasets to build a clear baseline for all local human activity: settlements, tourism infrastructure, and general infrastructure). The files are ready now.

Source and Acquisition

I grabbed all three datasets from OpenStreetMap. (So I ran specific Overpass Turbo queries filtered exactly to each island's borders, which gave me one raw GeoJSON export per island for each distinct category to create fifteen total files). So I chose OSM because it was practical. It gives me consistent, free data across five different national administrative systems—Maldives, India, Seychelles, Fiji, n Spain. (No single government office provides a dataset tht covers all five islands in the same format, so I had to use OSM to keep my framework uniform)..... This solved a huge matching problem.

The Column Bloat Problem

My raw OSM exports had an absurd number of columns. (Usually, I found around 200 to 240 columns in every single file, but most of it was total garbage like place name translations into dozens of languages such as name:ta, name:zh Latn pinyin, or wikipedia:ur, along with Wikidata links and contact info tht had nothing to do with settlement identity, population, or location). I did not want to carry this dead weight around. It made checking files by hand a massive headache.

So, I chose to clean all fifteen files the exact same way. Read them into my script using geopandas nd trimmed down the data to columns that actually mattered..... (For settlements, I kept name, place type, population, nd geometry; for tourism, I chose name, tourism type, amenity type, and geometry; n for infrastructure, I saved

name, man made feature type, amenity type, power infrastructure type, nd geometry). Then I saved them as new `_clean.gpkg` files. (So I left my original raw GeoJSON files untouched so I did not lose any source data). My settlement files dropped from 240 columns down to jst 4 or 5. I did not lose a single row or any useful field during this process.

Bug: Sea Features Mislabeled as Settlements

So I found a weird bug while checking my clean settlement files. (My bounding boxes for the Maldives nd Lakshadweep data looked way too massive nd extended way outside the actual island borders). Two specific rows caused this mess in both files. They were labeled "Arabian Sea" n "Laccadive Sea." (These entries were obviously not real human settlements at all, but random OSM features for large water bodies that got accidentally dragged into my Overpass Turbo place type query because online contributors tag things inconsistently). It happens a lot on OSM.

Caught this mistake by doing a quick trick. Calculated the centroid coordinates for every single settlement, sorted everything by longitude, n checked the extreme values at both ends. (The two sea name rows stuck out like a sore thumb immediately because their coordinates were sitting far away from the actual settlement clusters). So, I chose to delete both rows directly from my cleaned `.gpkg` files. (My Maldives file dropped from 998 to 996 rows, nd my Lakshadweep data went from 38 down to 36 rows). This fixed the bug completely without making me re pull anything from OpenStreetMap.

Bug: Invalid Geometry in Tourism Data

I hit a major code crash. (While I was trying to read the Canary Islands tourism dataset, `geopandas.read_file()` suddenly threw a bad `GEOSException` error about an invalid number of points in a `LinearRing`). The file only had 2 points instead of the 4 pairs needed to make a proper loop. This is a really common mistake. (People who edit crowd sourced OpenStreetMap data frequently make these broken, half finished polygon lines by mistake).

So I fixed the crash easily. I chose to pass `on_invalid="ignore"` right inside the `geopandas.read_file()` command so my script drops unreadable geometries instead of dying. Then I ran an explicit check. (I wanted to count if there were any null geometries left after the read, and I found exactly two invalid rows in the Canary tourism file). My rows dropped from 10,877 down to 10,875 valid ones. Made this exact setup a standard step for all my later vector cleaning scripts. (Broken polygon defects can show up anywhere when u use open source OSM files, so adding this check keeps my pipeline safe). This saved my code from breaking again.

Duplicate and Naming Cleanup

I found a few naming inconsistencies here. (This issue became a recurring pattern throughout my whole project becuase manually re running exports keeps producing near duplicate files under slightly different names). So I had to double check everything. Fiji's protected area adjacent file existed twice in my directory as `FIJI_INFRAA_SLR.gpkg` with a silly typo extra "A" n as the clean `FIJI_INFRA_SLR.gpkg` version. They were identical in size. (Chose to delete the typo'd copy directly from my folder to save space). I used this same handling pattern every single time this issue popped up anywhere else. So I always checked the file sizes or row counts 1st to confirm whether the copies actually differ. (Then I deleted the confirmed duplicates immediately rather than leaving ambiguous near identical files sitting around making my folders look messy). This kept my directory clean.

Final Status of This Phase

Dataset	Islands Covered	Typical Column Reduction
Settlements	5 (2 outlier rows removed from 2 islands)	~240 → 4–5
Tourism	5 (2 invalid rows removed from 1 island)	~200–490 → 4
Infrastructure	5	~200 → 4

All fifteen files confirmed clean, correctly bounded, and free of invalid geometries by the end of this phase — the human activity baseline used through the rest of the project, including the encroachment analysis much later.

Entry 4

Status

Finished this part completely. (My log entry here covers how I collected and cleaned the three ecosystem buffer datasets that I need for my main hypothesis testing: mangrove extent, coral reef extent, and protected areas). All three sets are done.

Two Different Source Types

My ecosystem buffer data was not like my settlement data. (While I got all settlement, tourism, and infrastructure layers uniformly from OSM for all five islands, the ecosystem buffer data came from two completely different source structures depending on the island, which forced me to write two separate cleaning pipelines instead of using one shared script). This doubled my processing work.

Found that Fiji, Lakshadweep, and Seychelles mangrove and coral data came straight from WCMC's Global Mangrove Watch and coral reef atlases. (You can identify these files right away because they use a loud uppercase column convention like LAYER_NAME, METADATA_I, NAME, FAMILY, GENUS, SPECIES, DATA_TYPE, GIS_AREA_K, and REP_AREA_K). But the data for the Maldives and Canary Islands came from OSM. It used the exact same lowercase place, natural, and tourism schema that I found in my settlement files. (So I had to make sure every mangrove and coral cleaning script could detect which source format a given island was using to apply the correct column selection logic). Had to create a specific WCMC_KEEP_COLUMNS list for the 1st group and a separate OSM style list for the 2nd group instead of using one universal set. This kept my dataset neat.

Bug: The Maldives "Coral" File Wasn't Coral Data

So I found a huge data quality problem here. (When I 1st opened the coral_maldives.gpkg file, it had 4,230 rows and 79 columns, but looking at the column names made me realize a bunch of fields had absolutely nothing to do with real coral reefs). The columns were filled with random words like cuisine, spa, restaurant, rooms, payment:visa, payment:mastercard, opening_hours, and parking. This file was not a filtered coral dataset at all. (It was basically an unfiltered general OSM export for the whole Maldives region that someone had just named like it was coral specific data). It was a total mess.....

So, I chose to filter it down myself. I used the official OSM reef tagging convention by running `natural == "reef"` on the table. (This step correctly picked out 2,921 of the 4,230 rows as genuine coral reef polygons, meaning the other 1,309 rows were just unrelated tourism, hospitality, and amenity points that got mixed into

the same export by mistake). I cleaned it up completely. A reef attribute column with a single consistent value like "coral" across the correctly tagged rows proved to me tht my filter isolated the right feature type. (Doing this filtering step before any column cleanup ensured that my downstream analysis was not quietly ruined by thousands of irrelevant tourism records). This made my data clean.

Real Ecological Absences, Not Data Gaps

So I got zero results for two islands..... (When I was pulling specific ecosystem data, these two places came back completely empty, but I chose to investigate n found out they were genuine absences instead of data download failures). This is my standard rule. So I treat every single zero result as a real hypothesis to double check instead of jst hiding it like an error.

My Lakshadweep data showed zero mangrove features. This makes sense. (Mangroves are very rare in this specific archipelago because it is jst a small coral atoll chain tht lacks the muddy intertidal delta conditions tht mangrove trees actually need to grow). Accepted this as a true fact.....

So I also found zero coral reefs for the Canary Islands. (This fits its subtropical, volcanic, cooler water Atlantic setting perfectly because the island sits way outside the warm tropical shallow water temperature range tht corals need)..... I got independent cross confirmation for this later. Ran a separate Global Mangrove Watch bounding box query for another multi temporal analysis, n it came back totally empty for the Canary Islands even after I deliberately made the search box much wider. (This proved to me that this island's coastal ecology is genuinely different from the tropical islands, nd it was not a systematic acquisition problem with my download script). My data looks correct now.

Protected Areas: Swapping Out a Weak Source

So I hit an issue with protected areas. (For four islands—Maldives, Canary Islands, Seychelles, n Fiji—I originally got the protection dataset from OpenStreetMap using the QuickOSM plugin, but on checking I found an implausibly small number of features like just one protected area for Fiji n four for Maldives). This data felt completely wrong. Knew these regions hve plenty of officially gazetted marine n terrestrial protected areas, so it was jst bad coverage.

So, I switched my source. I chose to use the World Database on Protected Areas instead because it is the main global dataset from UNEP WCMC. The full WDPA download is massive at abt 4.15 gigabytes nd comes split into three sequentially numbered zip archives due to shapefile attribute table size limits. (So I unzipped everything nd filtered each of the three polygon shapefiles per island using the bbox parameter inside `geopandas.read_file()` so my script only read the relevant subset off the disk instead of loading the whole multi gigabyte global file into my computer memory). This is my standard memory efficient approach. Finally, I combined the results from all three archive parts into a single clean file for each island.

My feature counts jumped up a lot. (After I fixed my data source, the Maldives count went from 4 to 93, Fiji jumped from 1 to 126 because I combined two sub queries to cover its territory tht straddles the antimeridian, Seychelles grew from 4 to 49, n the Canary Islands went from 258 up to 385 features)..... This proved my old data was bad. I found tht Lakshadweep had zero WDPA registered protected areas..... (This matched my earlier zero result from OpenStreetMap perfectly, giving me solid cross source proof tht this is a real fact abt Lakshadweep instead of some random database glitch). So, I deleted the old bad files n replaced them with the new WDPA data across all four islands.

The raw WDPA entries had 34 columns each. (They were carrying a ton of heavy metadata about governance types, ownership types, management authorities, IUCN categories, n verification status tht I did not need at all). Chose to trim them down. So I kept only nine useful columns: name, designation, designation type, IUCN category, legal status, status year, reported area, nd GIS calculated area. (This is enough information for me to calculate protection type nd size without dragging along useless admin rows tht nothing in my project code actually calls). My final files look much smaller now.

Final Status of This Phase

Dataset	Islands With Data	Confirmed Genuine Absences
Mangroves	Maldives, Seychelles, Fiji, Canary Islands	Lakshadweep
Coral Reefs	Maldives, Fiji, Lakshadweep, Seychelles	Canary Islands
Protected Areas (WDPA)	Maldives, Seychelles, Fiji, Canary Islands	Lakshadweep

This phase set the ecosystem buffer baseline used both for the project's static ecosystem context n, later, as the foundation for the full multi temporal analysis in Entry 6.

Entry 5

Status

I finished this part completely. (My logs here cover the remaining baseline files I need for the project: height nd slope terrain maps, population density, nd old historical cyclone tracks). Everything is processed now.

Terrain: Elevation nd Slope

I got my elevation data from Copernicus DEM. (Luckily, the files for all five islands arrived in the correct EPSG:4326 coordinate system right from the start, so I did not have to waste time doing any map re projections there). That was an easy win.

But the slope files were annoying. So I found tht three islands—Canary, Fiji, n Lakshadweep—arrived in the wrong EPSG:3857 map system instead of my project baseline EPSG:4326. (I had to fix this by writing a script with rasterio's calculate_default_transform n reproject functions using bilinear resampling because tht is the correct way to handle smooth continuous elevation stuff instead of nearest neighbor resampling which is only for category grids like land cover). Seychelles was even worse because it had zero slope files in the folder. (Instead of crying or downloading new files, I chose to generate the slope layer directly from my already checked Seychelles elevation raster using GDAL's gdaldem slope command in my terminal). This gave me a perfectly matching slope raster without downloading any extra junk. It saved my day.

So I cross checked elevation values manually. (Matched my raster data against real geography maps as a sanity check because I did not want to jst blindly trust random numbers on my screen). My Canary Islands file maxed out at around 3,696 meters. (This lines up nicely with Mount Teide, which is Spain's highest peak sitting at roughly 3,715 meters). The Seychelles data hit a top peak of around 904 meters. (This fits its steep granite mountain geography like Morne Seychellois perfectly). On the other hand, the Maldives and Lakshadweep layers stayed strictly in a super low 0 to 20 meters range. (This matches exactly wht you expect for low lying

coral atoll nations on any map). This quick matching test gave me high confidence. My height layers were processed correctly and did not get silently corrupted or shifted.

Population: Checked Against Real Numbers

So I downloaded the 2020 WorldPop density rasters. (So I managed to get good data for four out of my five targets: Fiji, Seychelles, Maldives, n the Canary Islands, where I clipped the last one directly from Spain's big national population map). This part went smooth. But Lakshadweep did not get any population data in this phase. (The only online download option I found meant pulling a massive multi gigabyte raster covering the entire country of India, which felt totally stupid nd impractical for the tiny island area I actually needed). So I skipped it for now. Chose to put it as a low priority task for later.

So I validated each raster by myself. (I chose to sum up every single pixel value in the files to calculate a total population estimate, then I checked those sums directly against real world figures instead of jst assuming the data was correct because the code ran fine). My verification showed good numbers. The Canary Islands file gave me 2.09 million against the actual 2.2 million count. (Fiji gave me around 896,000 instead of 900,000, Seychelles calculated to 98,000 against the real 100,000, nd the Maldives raster hit 540,000 which is exactly inside its known real range). So I treated this step as highly essential. A file processing cleanly does not mean the data inside is correct.

Then my code hit a wall. (My biggest population file for Fiji crashed the script with a `rasterio.errors.RasterioIOError` because it could not allocate 884,736,000 bytes during a full array read). I had to change my approach..... I fixed this issue by reading the raster in small tiled blocks using `rasterio's block_windows` iterator instead of opening everything at once. (Accumulated the sum, minimum, nd maximum counts incrementally across the blocks, which is the exact same memory efficient pattern I use for all heavy files in this folder). This trick saved my system memory. Phew.

Cyclone Tracks: IBTrACS n a Wind Speed Consolidation Problem

I got historical cyclone track data from IBTrACS. (This is NOAA's standard global cyclone dataset that mixes reports from more than a dozen different meteorological offices). It was a very big file. I chose to filter down the full global shapefile per island using the exact same bounding box `geopandas.read_file(bbox=...)` approach tht I already used for my WDPA files. This method saved my computer from crashing.

Four out of my five islands returned real storm data. (I found that Fiji had 143 tracks, Lakshadweep had 4, Maldives had 5, and Seychelles had 2 entries in the final count). These numbers matched my expectations. But the Canary Islands file came back with absolutely zero rows. (Checked this specific place twice under two separately named filtering attempts in my script, but it is a true result because the Canary Islands sit entirely outside the typical Atlantic tropical cyclone formation n travel corridor). I logged it as a real geographical finding.

So I found another bad data quality issue here. (The wind speed column I wanted to use, called `WMO_WIND`, was completely blank for most of my records since it only had data for 1 out of 5 storms in the Maldives nd 31 out of 143 for Fiji). This was a total block. But I noticed tht IBTrACS separately logs wind estimates from different local national agencies like `USA_WIND`, `REU_WIND`, `BOM_WIND`, `TOK_WIND`, `CMA_WIND`, n `HKO_WIND`. So, I chose to use `pandas' bfill(axis=1)` trick across a prioritized list of these agency columns. (This smart code script automatically grabs the very 1st available non null wind value for each storm row no matter which country's office tracked it originally). It worked like magic. Managed to improve Fiji's useful wind speed data from 31 up to 87 out of 143 rows. My data counts for the other three small islands—Maldives, Seychelles,

nd Lakshadweep—stayed low, but tht is jst because those places hve a real natural scarcity of storms rather than a bug in my coding consolidation method. (Confirmed this fact by running the exact same multi agency check script against those islands nd found absolutely nothing extra beyond wht my 1st four agency check caught). My logic was solid.

So I saved this new consolidated wind speed field into a clean file. (So I trimmed down the huge raw IBTrACS table which had over 150 useless columns down to jst a small set containing storm ID, season, name, timestamp, nature, position, consolidated wind speed, n distance to land). I saved everything as a new _final.gpkg file for each island. This new file replaced the bulky raw file entirely for my next steps.

Final Status of This Phase

Dataset	Islands With Full Coverage	Notes
Elevation	All 5	Cross checked against known peak elevations
Slope	All 5	3 islands reprojected; Seychelles generated from elevation
Population	4 of 5	Lakshadweep deferred; all 4 checked against real world totals
Cyclone Tracks	4 of 5	Canary Islands genuinely absent; wind speed consolidated across agencies

This phase closed out the full baseline dataset inventory needed before starting the project's core multi temporal nd cross dataset analyses, covered nxt.

Entry 6

Status

So I finished this main chunk completely. (My logs here cover the core analytical part of the project where I calculate actual multi temporal ecosystem trend files instead of jst relying on lazy, single before nd after snapshots to measure physical danger). This is the biggest part.

Why Multi Temporal

My old data from Entry 4 only showed a single snapshot in time. (So I realized tht if I really want to test whether mangroves or coral reefs are rotting away, I absolutely need dataset layers recorded across multiple, well separated years). A single comparison is a bad idea. (It is too easy for a simple before n after check to look fake because of random noise or an unusual reference starting period). My data needs to be bulletproof. A trend finding tht holds true across several independent time blocks on a timeline is simply a lot more convincing to examiners.

Mangrove Time Series: Three Points

I used the Global Mangrove Watch archive..... (The GMW archive gives you pre processed, ready to use annual mangrove extent snapshots so I did not hve to waste my hours sitting n calculating the boundary limits from raw satellite pictures by hand). It saved my energy. I chose three specific time points for my project: 1996

which is the earliest baseline available, 2010 for the exact midpoint, nd 2020 as the most recent single year data block. (This covers a proper 24-year span with a clear check right in the middle because a single big 24-year jump completely hides whether the jungle loss was gradual or jst happened suddenly in a tiny window). I wanted to see the full timeline clearly.

Noticed a small trick inside the GMW folder. (The database provides files both as single year snapshots like `gmw_v3_2020_vec.zip` n as change detection vectors spanning a date range like `gmw_v3_f1996_t2020_vec.zip` where f means from nd t means to). So I had to make a choice. I chose to independently derive the total area change by my own scripts using the three separately verified snapshots. (So I did not want to blindly lean on GMW's own pre computed change detection algorithm, so I downloaded n ran the three single year files instead of touching their mixed pre computed change file). This choice kept my calculations 100% transparent.

I filtered each global GMW shapefile. (The raw files are massive at around 894 to 948 megabytes each, so I used the exact same bbox based approach from my WDPA and IBTrACS scripts to avoid loading the whole global file into my laptop memory). This kept my machine from slowing down. So I generated clean per island nd per year mangrove extent files from this step. (Only did this for the four islands where mangrove trees actually grow because my data check in Entry 4 already confirmed tht Lakshadweep has zero mangroves). My folders are fully organized now.

Bug: Feature Count Isn't a Valid Area Proxy

So I ran an early data check. (I used raw polygon feature counts across the three time points as a quick way to see changes, n it looked like Fiji had a massive drop because it went from 5,608 features in 1996 down to 4,810 in 2020 which looks like a 14.2% loss). Did not trust this number. I knew that feature counts nd actual mapped areas are completely different things on a map. (For example, one large continuous mangrove forest can show up as a single polygon in one year, but it can easily break into several small adjacent polygons in another year purely because of how the satellite classification algorithm decides to slice that specific imagery). There might be zero real area change. Had to write a better test.

I tested this issue directly. (So I reprojected all three years for each of the three mangrove islands into an equal area CRS called EPSG:6933 because it is appropriate for real area calculations unlike standard geographic coordinates which badly distort shapes depending on ur latitude). Then I summed the total area in square kilometers instead of using polygon counts. So I got a totally different, more reliable result. (Fiji's mangrove area was basically stable across all three points sitting at 485.72 square kilometers in 1996, then 487.97 in 2010, n 488.41 in 2020 which means a small net increase of just 0.6%). My calculations for the Maldives nd Seychelles also showed less than 0.01 square kilometers of real change over the full 24-year span. So the fake decline from feature counts was jst a segmentation artifact between different years' imagery. The actual mangrove extent held essentially stable across all three tested islands over nearly a quarter century, which directly changed how I framed my hypothesis testing later in the project log. This step saved me from making a big mistake.

Coral Reef Condition: A Thermal Stress Time Series

Coral degradation shows up as bleaching. (Basically, it is a health nd condition change coming from heavy thermal stress instead of a simple drop in the mapped physical boundary line, so I absolutely needed a totally different data source nd metric than my mangrove tracking setup). My 1st plan failed. So I considered using the Allen Coral Atlas 1st, but it turned out to be jst a single snapshot composite built from 2018 to 2020 pictures instead of a real multi year timeline layer. It did not fit my trend requirement.

So, I chose a better tool. (Found tht NOAA Coral Reef Watch's Degree Heating Week product was the perfect match because it gives a continuous daily satellite derived measure of accumulated heat stress from 1985 up to the present day). This dataset tracks everything nicely. (I konw that values above 4°C weeks cause major coral bleaching n numbers above 8°C weeks mean severe bleaching n dead corals)..... This metric measures the actual biological stress driver directly. It makes complete sense because hard coral reef structures can physically stand there in the ocean for years after a bad heating event even when the living animals inside are already dead. My analytical logic was solid here.

Bug: Access Denied, Then the Wrong Variable Name

I accessed NOAA's ERDDAP server. (I sent constructed URL requests directly to their system instead of clicking links to download files manually, which matches the API based coding patterns I use everywhere else in my project folder). My 1st code attempt failed. The server blocked me with a nasty HTTP 403 Forbidden error because it was rejecting Python's default requests user agent string as an automated bot script. I fixed this issue quickly. (Explicitly set a real browser style User Agent header line using Mozilla/5.0 (Windows NT 10.0; Win64; x64) inside my script, after which the server finally started accepting my connections). It returned a 500 error instead of a 403. My access was granted, but I had a different problem left to solve.

The 500 code gave me a clear hint. (Teh exact error string destinationVariableName=CRW_DHW wasn't found in datasetID=noaacrwdhwDaily meant I had written down the wrong variable name inside my script request block)..... So I had to look it up..... Checked the dataset's actual online metadata webpage directly with my browser and found the correct column name, degree_heating_week. (I swapped this proper string right back into my Python code, n it fixed my API request completely). My data started downloading nicely.

Bug: A Coordinate That Landed on Solid Ground

Four islands worked perfectly right away. (Got valid, non null DHW series numbers for them using representative offshore coordinates that I picked out from my map grid). But my Canary Islands data failed. The 1st coordinate point I chose came back entirely empty because 305 out of 305 values were totally null. (I traced this bug nd found my coordinate was falling exactly on the landmass or way too close to a beach pixel in their 5km resolution grid, which happens easily because the Canary Islands are very small). It was a frustrating block.

Chose to test three alternative offshore coordinates instead of wasting time. (I ran a quick single month request for each new candidate point instead of lazily repeating a full 24-year download for every single location). This smart shortcut worked. One point sitting just south of Gran Canaria returned a valid value. (I immediately used this working coordinate to re pull the full Canary Islands time series successfully without any errors). My coral data is fully complete now.

DHW Trend Result

I compared two different time blocks. (So I took an early reference period from 1996 to 2000 n calculated its average against a recent period spanning 2016 to 2020 to check for changes)..... My results showed a clear pattern. So I found tht four islands had rising thermal stress numbers..... (My calculations showed the Maldives went up by +0.17°C weeks, Fiji hit +0.10, Lakshadweep got +0.08, and the Seychelles jumped up by a massive +0.68 with a single max reading of 10.47°C weeks which sits right in the dangerous range for severe bleaching n coral death). This Seychelles number is very scary.

But my Canary Islands data looked totally different. (It actually showed a slight decrease of -0.05 , making it the only island out of my full group of five that was not trending upward on the heat charts)..... I checked why this happened. (I realized this fits perfectly with its distinct Atlantic climate regime because it operates completely differently compared to the other four Indian Ocean n Pacific islands in my folder). My data mapping looks ready now.

Physical Exposure: Settlement Level Sea Level Rise Risk

So I calculated the physical exposure numbers nxt. (I chose to sample my elevation raster right at each settlement's center point using rasterio's sample method across all five islands so I could find the percentage of villages sitting at or below the one meter sea level rise mark). This gave me a very clear picture. (My data showed a massive geographical spread where 99.1% of Maldives settlements are in high danger, followed by 78.3% for Seychelles, 77.8% for Lakshadweep, 32.0% for Fiji, nd just 12.1% for the Canary Islands). The gap between them is huge. (These results directly show the big difference between flat coral atoll islands nd tall volcanic mountain grounds tht I already saw during my height checks back in Entry 5). My code script ran perfectly.

Compound Vulnerability Score

Mixed the two risk signals. (So I took my independently checked physical sea level rise exposure numbers nd my coral thermal stress trend data, scaled each of them onto a basic 0-1 range using min max across the five islands, n simply averaged them with equal weighting). This was a major calculation step. So I chose to leave the mangrove trend out of this formula entirely. (My long term data analysis did not find any real area loss for mangroves, so there was basically no measurable drop to weight in anyway). This made my logic very clear.

My final list ranking came out. I found that Seychelles scored highest at 0.880, followed by Maldives at 0.651, Lakshadweep at 0.467, Fiji at 0.217, and the Canary Islands sitting at 0.000 as the reference minimum by construction..... This proves my project's main argument. (See, the Maldives has a much higher physical risk by itself with 99.1% versus Seychelles' 78.3%, but Seychelles wins the highest total danger spot once u actually mix in the bad coral reef dying trends). Looking at water lines alone is a trap. (If I only looked at physical exposure, I would hve picked the completely wrong island as the most dangerous zone). My composite approach solved this issue.

Final Status of This Phase

Analysis	Result
Mangrove extent (1996/2010/2020)	Stable across all 3 tested islands; no decline detected
Coral thermal stress (1996–2020)	Increasing in 4 of 5 islands; Seychelles most severe
Physical SLR exposure	Wide variation, 12.1%–99.1% of settlements at risk
Compound Vulnerability Score	Seychelles ranked highest; Canary Islands lowest

Entry 7

Status

Finished the two final analytical pieces..... (For this last step, I ran statistical tests to see if official protected area lines actually match up with my verified risk scores, and then I checked whether local villages are encroaching right into the fragile ecosystem buffer zones over multiple decades). So I also put together a complete synthesis of the whole project log. Everything is fully done now.

Adding a Governance Dimension

I finished my main risk analysis. (I chose to add a deliberate extra phase right here because I really wanted to connect my work to real policy evaluation instead of leaving it like a basic, boring science project). I had a plan. Instead of treating protected area boundaries as jst some unexamined backdrop maps, I formulated a specific testable hypothesis. (So I think tht official protection marks like WDPA coverage should go up in places where my Compound Vulnerability Score from Entry 6 is high, if the governments are actually listening to data). So I will find the truth.

Bug: Comparing Land Area to Marine EEZs

My 1st data test failed completely..... (Tried to calculate a simple protected area to land area ratio, but my script gave me a totally nonsense result of 1005.69 for the Seychelles which basically means the protected zone size went over the actual island land size by more than a thousand times). It was a ridiculous output. Traced this bug down into the code. (Found tht the massive WDPA dataset accidentally included the entire Seychelles Exclusive Economic Zone because my simple bounding box filter from Entry 4 swept up over a million square kilometers of open ocean along with the real land based parks). That marine area is huge. Comparing tht gigantic sea number against a tiny island land area denominator was nvr going to give me any real, readable results.

So I fixed this error quickly. (So I restricted my protected area calculation to a strict 10km coastal buffer around each island's boundary instead of using the full captured WDPA extent because a settlement's actual exposure is tied to nearby coastal protection nd not to some random offshore marine park hundreds of kilometers away). This made much more sense. Implemented this by buffering each island's boundary geometry by 10km after reprojecting to an equal area CRS. (Then I intersected tht fresh buffer zone with each island's WDPA polygons before summing the total area). Much better ratios came out..... Seychelles dropped from 1005.69 down to a clean 11.32. (It is still the highest value of my five islands but it is no longer totally absurd, while my Maldives score hit 3.90, Canary Islands got 2.32, Fiji sat at 0.42, n Lakshadweep landed on 0.00 which matches its verified zero coverage from Entry 4). My numbers are fully sorted now.....

Governance Alignment Result

So I ran a Pearson correlation test next. (So I took my Compound Vulnerability Score for each island and checked it against its coastal buffer WDPA ratio using the `scipy.stats.pearsonr` function in my Python script). The output gave me $r=0.727$. This means I found a moderately strong positive relationship. (See, my data shows tht Seychelles has the highest vulnerability nd also carries the highest protection ratio, which looks like a good pattern). But my p value came out to 0.164. (This means my test missed the conventional 0.05 statistical threshold completely so it is not 100% solid proof). I reported this honestly in my log as suggestive rather than confirmatory. The direction definitely looks like risk responsive governance. But the main headache is tht five islands jst do not give enough statistical power to be fully confident abt it, which is the exact same small sample size limitation tht ruined a comparably small country sample earlier in my old project work. I cannot change this reality.

Bug: A NoData Artifact Hiding Inside My Canary Islands Number

Went back n double checked something tht had been bugging me a little. (My Canary Islands physical exposure number of 12.1% from Entry 6 jst felt off compared to how flat nd low lying its elevation profile looked in my earlier height checks, so I decided to open up the raw per point raster sample results instead of jst trusting the summary percentage). Good thing I did tht. (Dug through the point by point output n found tht 649 out of my full 5,483 raster sample points for the Canary Islands were actually NoData values sitting right at the coastline edge of my raster grid, but my earlier script was silently counting a chunk of those NoData points as if they were real, valid at risk settlement points instead of throwing them out first). Tht is a real bug, not jst noise in the numbers.

Fixed it by properly excluding those 649 NoData points before doing the percentage math. (Once I recalculated using only the genuinely valid raster samples, the Canary Islands physical exposure number dropped all the way down from 12.1% to a much more realistic 0.3%, which actually fits its flat volcanic coastline way better than tht old inflated figure ever did)..... Phew. Because this feeds directly into my Compound Vulnerability Score formula from earlier in this entry, I had to rerun tht whole calculation too. My corrected scores came out as Seychelles at 0.895, Maldives at 0.651, Lakshadweep at 0.481, nd Fiji at 0.263, with the Canary Islands still sitting at 0.000 as the reference minimum (the ranking order stays exactly the same, Seychelles is still on top, jst the individual numbers shifted a little bit). This also meant redoing my Pearson correlation test right above since it leans on these same vulnerability scores. (My corrected governance alignment result came out as $r=0.718$ nd $p=0.172$, which is basically the same moderately strong, not quite statistically significant pattern as before, jst slightly adjusted). Nice tht my overall story n ranking didn't change at all, but I'm really glad I caught this before locking anything in as final 😊

Settlement Encroachment: Satellite Derived Built Up Area Change

Coded the last analytical part nxt. (Chose to test if new houses n infrastructure expansions are crowding right inside or close to the fragile ecosystem buffer zones over a long time timeline by using the Normalized Difference Built up Index). This is called NDBI for short. (So I built it the exact same way people make standard NDVI but I tuned my code parameters to detect built up urban surfaces instead of green trees nd grass by calculating values directly from Sentinel-2 shortwave infrared and near infrared bands). It ran nicely. Applied this index math strictly to the three islands where mangroves are actually growing—Maldives, Seychelles, nd Fiji. (My script compared an old 2016 baseline satellite dataset directly against a recent 2024 time period to map out the new construction expansion). This gives me the final building numbers.

Bug: Empty Results, No Error

My 1st data pull failed quietly. (So I used a full calendar year aggregation window called P1Y for each island, which returned an HTTP 200 success code but gave me a totally empty data array like "data": [] nd a "geometryPixelCount": null field for every single request across all three islands nd both periods). This was an annoying silent failure. (So I had to read the raw JSON text directly to catch this bug because I couldn't just blindly trust the green request success status alone). It was tricky.....

Made two changes to fix it. Switched the aggregation window length from a full year down to six months using P6M. (I chose this because of a lesson from my older NO₂ nd NDVI data work where certain Sentinel Hub interval lengths behave inconsistently depending on your specific data type). Then I did one more step. I added an explicit maxCloudCoverage: 30 filter line to my request script. (Persistent empty values across a full year for tropical islands usually mean heavy cloud cover is blocking any usable clear sky satellite observations).

Both edits worked perfectly together. All six of my requests—meaning three islands multiplied by two time periods—came back fully populated with numbers.

Encroachment Result

Compared the average NDBI values. (Looked closely at the data between my two selected time blocks and found tht both Maldives went up by +0.0461 n Seychelles grew by +0.1135 to show a clear increase matching real house nd infrastructure expansion right near the ecosystem buffer zones between 2016 nd 2024)..... Seychelles showed the largest growth jump. (My data reveals it actually crossed completely from a negative average in 2016 which means vegetation dominated lands straight into a positive value in 2024 showing it is now built up dominated ground). This is a massive change. But Fiji showed basically zero shift sitting at -0.0008 on my calculator. (This dead flat trend lines up perfectly with the overall stability tht I already saw earlier when checking its mangrove extent dataset). My numbers match across different metrics.

Final Synthesis

Seychelles kept popping up everywhere. (Across every analytical piece I coded, this specific place showed the absolute strongest proof of a compounding, self reinforcing vulnerability pattern because it carried the highest Compound Vulnerability Score driven by the worst coral thermal stress trends). My data points in one clear direction. It also has the highest coastal protected area ratio. (Even though I could not statistically confirm this governance response with my small sample size, the policy direction looks correct). It also gave me the biggest house building encroachment signal.

Fiji was the opposite story. (My data shows it stayed completely stable across every single temporal measure over the decades with flat mangrove boundaries, the smallest coral heat stress increase among all working islands, n zero infrastructure encroachment). This was an amazing comparison. It serves as a genuinely low change reference case right against Seychelles’ crazy high change profile in my sheets. My full synthesis is complete. My mangrove decline hypothesis completely failed. (I tested H2 with the exact same strict code rigor n multi point timeline validation as everything else in my project folder, but the actual evidence did not support my guess in any of the three islands where trees grow). So I found no real jungle loss. (Writing this down honestly in my log instead of tweaking my code or hiding the bad results feels like a solid, helpful contribution). (My data clearly shows that nature buffer degradation is not uniform across different ecosystem types at all because coral reefs in my five island sample are measurably more vulnerable to heat stress than mangroves over this multi decadal timeline). This is a big deal for resource prioritization. (It tells global adaptation n conservation finance groups exactly where to spend their budget 1st instead of wasting money on stable zones). My final log sheets are sorted.

Complete Project Data Inventory

Category	Coverage
Settlements, Tourism, Infrastructure	5 islands each
Mangroves (multi temporal, 3 time points)	4 islands (Lakshadweep genuinely absent)
Coral Reefs (extent)	4 islands (Canary Islands genuinely absent)
Coral Thermal Stress (24-year time series)	5 islands
Protected Areas (WDPA)	4 islands (Lakshadweep genuinely absent)

Category	Coverage
Elevation, Slope	5 islands each
Population	4 islands (Lakshadweep deferred)
Cyclone Tracks	4 islands (Canary Islands genuinely absent)
Settlement Encroachment (NDBI)	3 islands (mangrove present islands only)

Investigated every single zero result. (Whenever I hit an unexpected finding along the way—like Lakshadweep's completely absent mangroves, the Canary Islands' absent corals and cyclones, or my mangrove decline hypothesis not holding up—I chose to dig deep n confirmed them as genuine geographical facts or traced them back to a specific, fixable data cause). I never assumed these weird numbers were just random bugs or quietly worked around them to make my life easy..... (Where my original hypothesis jst did not pan out in the stats, I wrote it down openly as a real, valuable finding in my log sheets instead of tricking the system or reframing the txt to fit my original baseline expectations).

Entry 8

Status

Added a final part here. (So I chose to check cyclone driven satellite damage detection to get supporting evidence for my risk numbers, but I hit a very annoying data availability constraint while working on the script). It was a surprise block. Wanted to see if the satellite imagery could prove the storm damage on the ground immediately. (But my data search showed tht finding clean, cloud free images right after a major cyclone is nearly impossible for these tiny tropical islands). This limited my calculation steps. So I logged this asset gap as a real project constraint for later.

Context n Why It Got Added

My core analysis centers on sea level rise and ecosystem buffer degradation. (Made a deliberate scope call to treat cyclone activity as a background factor rather than a primary pillar because data from an earlier, differently framed version of this island nations project showed meaningful cyclone intensity variation only for Fiji, with the other four islands' historical storm records being too weak and low intensity to detect a genuine damage signal on a satellite map). This earlier version was originally built around cyclone impacts specifically. So I changed my mind.

So I did not want to throw away tht earlier cyclone related work once my project's framing shifted toward water lines nd coral death. (I chose to keep it as a smaller, clearly scoped supporting addition inside my folders instead of messing up nd re expanding my core analytical structure). This saved a lot of code. Two Sentinel-2 NDVI before and after comparisons from tht older project phase were still sitting right inside my data folder..... (These maps cover Fiji's two strongest historically recorded cyclones, Winston in 2016 n Yasa in 2020, n they stayed completely unaffected by my later project rename). So I am glad I saved them 😊

Trying to Extend This to the Other Four Islands

I tried an extra test 1st. (Before I finally settled on Fiji as my only island with clear cyclone damage evidence, I tried extending the exact same satellite damage detection approach to the Maldives, Seychelles, n Lakshadweep using each island's single strongest recorded storm from the IBTrACS database tht I already set up in Entry 5). This took some extra time. 1st, I had to pin down each island's peak intensity event name nd its exact calendar date. (I needed to do this step because building a proper before nd after satellite imagery window completely depends on knowing the exact day when the storm actually hit the beach). This step was very necessary for my scripts.

Ran Into a Satellite Record Start Date Wall

I hit a major database wall while writing my acquisition script. (Found out a huge fact tht nvr came up before this step: Sentinel-2, which is the exact satellite I use for all my green NDVI maps in this project folder, only launched in the year 2015, meaning zero pictures exist before tht date). It was a total shocker. Then I checked the historical dates for my three candidate islands' strongest recorded cyclones against this satellite launch cutoff line. (My data matching showed tht all three storms happened way before Sentinel-2 was even built: the Maldives' biggest cyclone hit back in 2006, Seychelles faced its worst one in 1996, nd Lakshadweep had its massive peak storm in 1997)..... This made things impossible. No valid "before" satellite imagery could evr exist for any of these islands no matter how smartly I chose to code the API requests. The satellite hardware was simply not flying around in space yet during those old years. This data scarcity completely killed tht line of analysis.

I built this check directly into my script. (I chose to add a strict pre condition logic line tht checks the cyclone year against a 2016 cutoff rule before my code attempts to run any image download requests). This saved me from extra debugging work. (Instead of letting my code run blindly and return an uninformative empty response that would need separate manual diagnosis later, the code stops early if the dates are too old). So I use this exact upfront constraint pattern everywhere else in my project folder. Running my code confirmed everything worked. (My script skipped all three islands immediately n printed out an explicit, clear reason on my terminal screen for each skipped folder). It handled the gaps perfectly.

How This Got Reported

I did not treat this as a gap to paper over. (For example, I refused to swap in a weaker, later cyclone for one of the affected islands jst to trick my scripts into producing some fake results). It is a genuine structural limit of satellite based verification..... (Any remote sensing method u try is strictly bounded by when its satellite hardware platform actually started operating in space, n events before tht date cannot be verified by that method no matter how smartly u configure the API request parameters). This is a fixed reality. I documented it transparently as a scope limit on this one supplementary analysis block specifically. (The best part is that it does not touch my project's core sea level rise or ecosystem degradation findings because those calculations do not depend on Sentinel-2's coverage window the same way at all). My primary results stay fully safe.

Final Cyclone Damage Proxy Status

Island	Cyclone Damage Evidence	Reason
Fiji	Available (Winston 2016, Yasa 2020)	Both events post date Sentinel-2 availability
Maldives	Not available	Strongest recorded cyclone (2006) predates Sentinel-2

Island	Cyclone Damage Evidence	Reason
Seychelles	Not available	Strongest recorded cyclone (1996) predates Sentinel-2
Lakshadweep	Not available	Strongest recorded cyclone (1997) predates Sentinel-2
Canary Islands	Not applicable	No cyclone events recorded in this region (Entry 5)

Analyzed the two Fiji results next..... (My numbers showed a clear NDVI decline of -0.0246 after Cyclone Winston which hit at 150 knots as a Category 5-equivalent storm, versus zero clear decline sitting at $+0.0049$ after Cyclone Yasa which was weaker at 115 knots)..... These numbers serve as good supporting evidence. So I found that satellite detectable vegetation damage only shows up at extreme storm intensity in this sample. (This matches nicely with the broader ecosystem stability that I already saw inside Fiji's mangrove dataset earlier). I kept it framed exactly that way. Chose to report it strictly as one island's result instead of extrapolating it to the other four islands.

This specific case was a bit different. (Most of my earlier debugging work was about verifying data was correct once it had already been acquired by checking bounding boxes, validating population totals, or confirming real ecological absences). This time I faced a unique challenge..... I needed to catch a structural limit of the data acquisition method itself. (I had to check the satellite's fixed launch date before even attempting the acquisition loop instead of only finding out through an empty result after running the script). This quick trick saved my processing time.